

RBE 595 – HW1

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Part 2: Gripper Selection for Recycling Sorting

Object Characterization

The objects present on a conveyor belt contain significant material and geometric diversity. Some of them are as follows: glass bottles, aluminum cans (in varying shapes and sizes), rigid and flexible plastic items, cardboard and paper with distinct friction properties on their surfaces and rigidity. What makes this situation unique is that unlike the case of a structured pick-and-place task with known object models, items arrive in arbitrary, random poses and orientations on the moving belt. Additionally, surface condition varies as well: items may be wet, wrapped in residue, or structurally deformed which affects achievable contact and seal quality. Some items may also be fragile: glass requires controlled, limited-force contact to avoid breakage, while cans and rigid plastics tolerate higher grip forces. Finally, the conveyor imposes a throughput constraint: the gripper must grasp and release the objects quickly enough to keep pace with the speed of the moving belt.

Gripper Selection and Justification

Selection: **Universal jamming gripper**

Justification: Given the extreme shape and randomness of poses as discussed above, a universal jamming gripper is the most apt choice. A jamming gripper is able to conform passively to arbitrary object geometry on contact, then rigidifies via granular jamming, which eliminates the need for precise pre-grasping planning pipeline for each different object type: something that a simple parallel jaw gripper is unable to offer. This directly addresses the lack of consistent grasp affordances in an unstructured recycling stream. The universal jamming gripper also outperforms suction in this context as suction requires a flat and non-porous surface to form an airtight seal, which fails on cushion cans or items made of plastic which constitute the majority of irregular, damaged geometry in a recycling stream. The downside is that jamming grippers are generally slower to actuate than suction, and we may need to re-evaluate the grip force for very fragile items like glass. But given that reliability across an unpredictable, mixed-material stream is the dominant challenge here, that disadvantage is acceptable relative to the alternatives.